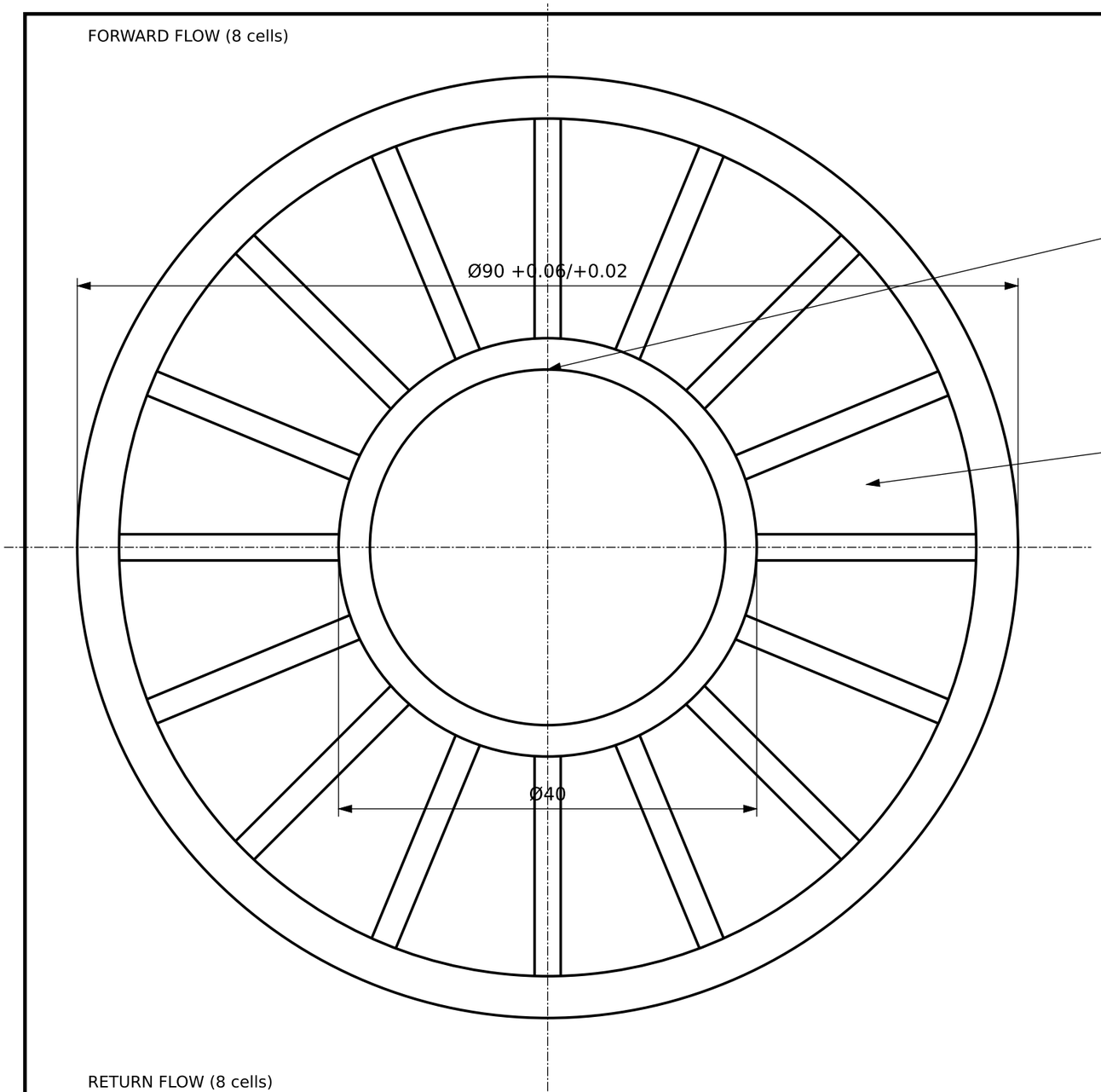

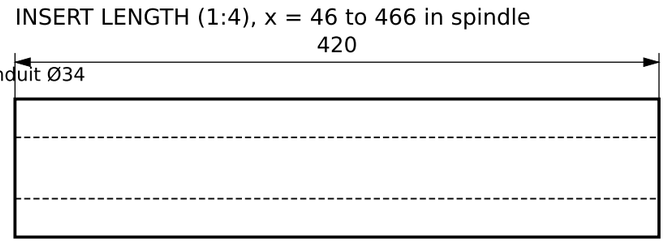

NOTES

1. The two fins at 0° and 180° are the flow partition: they are continuous and sealed to the front plenum cap so that air flows forward in the upper 8 cells and returns in the lower 8 (U-flow, both ports at the rear).
2. Front plenum cap (6082, 10 mm, O-ring to bore) closes the cells 10 mm from the insert front end and lets the Ø34 conduit pass through to the slip ring position.
3. Shrink into spindle bore with liquid nitrogen; interference 0.02 to 0.06 at 20 °C gives contact pressure in operation (Al expands more than steel). Thermal paste is not used (shrink fit).
4. Profile tolerances EN 755-9; cut length ±0.5; deburr; no anodising on the OD (thermal contact).
5. Rear manifold (fan, filter, inlet/outlet split) bolts to the spindle rear face, see NP1-300 and doc 14 section 6.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

SPINDLE COOLING INSERT (FINNED EXTRUSION)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-110	B · 2026-11-03	2:1 / 1:4
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
EN AW-6082-T6 extrusion	2.6 kg	Claude / not checked